

Si Microring Resonator Modulators at >200Gb/s

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Abstract: The progression of the development of all-silicon based microring resonator modulators for high bitrates is reviewed. Trade-offs and challenges are explored. © 2026 The Author(s)

1. Introduction

Silicon microring resonator modulators (MRMs) have been studied for more than two decades as a pathway to compact, CMOS-compatible optical modulation with high integration density [1,2]. Today, the device continues to attract interest as AI workloads scale rapidly and we shift from CPU-centric datacenters to GPU-driven systems that need large amounts of power and interconnect bandwidth. Co-Packaged Optics (CPO) is a compelling solution for these large-scale AI factories, offering up to $5\times$ better power efficiency and $10\times$ improved resiliency compared to traditional pluggable optics [3]. In these CPO architectures, the optical engine is mounted on the same substrate as the switch IC, shortening high-speed electrical paths and reducing failure-prone interfaces. MRMs are particularly attractive for CPO because of their ultra-compact footprint, enabling high bandwidth density, and CMOS-compatible manufacturing for 3D integration of photonic integrated circuit (PICs) and electrical integrated circuit (EICs). The announcement of the world's first 1.6 Tbps (200 Gbps per lane) CPO switch at GPU Technology Conference 2025, featuring MRMs fabricated using Taiwan Semiconductor Manufacturing Company's (TSMC) Compact Universal Photonic Engine (COUPE) process, represents a significant milestone in the evolution of silicon photonics and has reignited interest in advancing MRMs for even higher data rates in future generations [4,5]. This paper examines key silicon MRM design tradeoffs, theory and modeling advances, and reports experimental results at high bit rates.

2. Ring modulator modeling and analysis

Optical intensity modulator performance is dictated by a tightly coupled set of metrics—electro-optic (EO) bandwidth (BW), insertion loss (IL), linearity, modulated extinction ratio, and chirp—whose interdependence is particularly pronounced in MRMs. The behavior of the MRM at high optical power is also very important design consideration. Overall, performance depends on co-optimization of the modulator with the driver and package. Ultimately, performance depends on co-optimization of the modulator with the EIC driver and its path to the PIC, and benefits from electronic/photonic design automation co-simulation. Valuable intuition, however, can be obtained by examining the modulator's electrical drive network and the resonator's optical dynamics, which are described by circuit analysis and coupled-mode theory, respectively [6–11].

2.1. Electrical analysis

Electrically, the delivered modulation voltage over frequency is set by the junction resistance R_j , capacitance C_j , equivalent inductance, and the source and termination impedances. A common model of the circuit is shown in Fig. 1a, where the voltage transfer to C_j , $H_e(\omega) = V_j / V_{in}$, is to be optimized. Neglecting parasitics, the response is approximated by a single-pole RC, while inductive peaking produces a two-pole response. The 3-dB BW can be estimated using the expressions in Fig. 1b. A fundamental trade-off in silicon modulators exists between phase-shifter efficiency and C_j , as will be shown later, preventing achievement of a high-BW junction. Hence recent literature is pursuing forms of inductive peaking [9,12]. Reducing R_j by higher doping, is accompanied by an increase in optical loss. Both optical loss and junction resistance increase with local temperature, which can be substantial at high optical powers for a ring modulator.

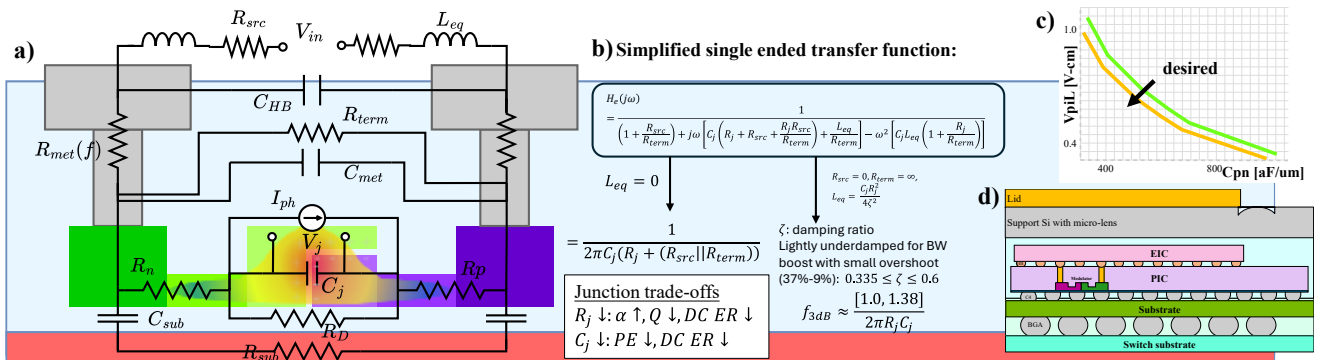


Fig. 1 a) Equivalent circuit model of a phase shifter, b) simple transfer function, c) silicon photonic trade-off with junction capacitance, d) 3D IC integration for low parasitic EIC to PIC connections.

The RC product is fixed with radius, but the optimum choice of inductance depends on the absolute values. Because the optimum peaking inductance can be relatively small, parasitic control is essential. Recent 3D integration of PICs and EICs by hybrid bonding (e.g., TSMC COUPE/SoIC®) technologies [5,13], as depicted in Fig. 1d, shorten the EIC–PIC interface, reducing interconnect RLC parasitics and enabling more controlled implementation of high-speed electrical passives. Developments in 3D IC electromagnetic simulation workflows (e.g., RaptorX, EMX, and PeakView®) further improve the accuracy of EIC–PIC interconnect parasitic extraction. The sensitivity to inductance also emphasizes careful RF de-embedding for accurate junction RC extraction and reporting.

2.2. Optical analysis

Optically, the resonator coupling condition and the laser-resonance detuning jointly determine the DC transfer curve and the dynamic (small- and large-signal) modulation response. The DC and AC response is asymmetric with respect to positive and negative detuning because of the junction's loss dependency on bias voltage. Due to optical self-heating, biasing the resonance is practically possible only on the lower wavelength side of the resonance at high optical powers. A summary of the ring's optical behavior is shown in Fig. 2, where the small signal response is described by a complex conjugate pair of poles and a zero [7]. The zero creates additional dependency on the coupling ratio of the MRM and the sign of the detuning.

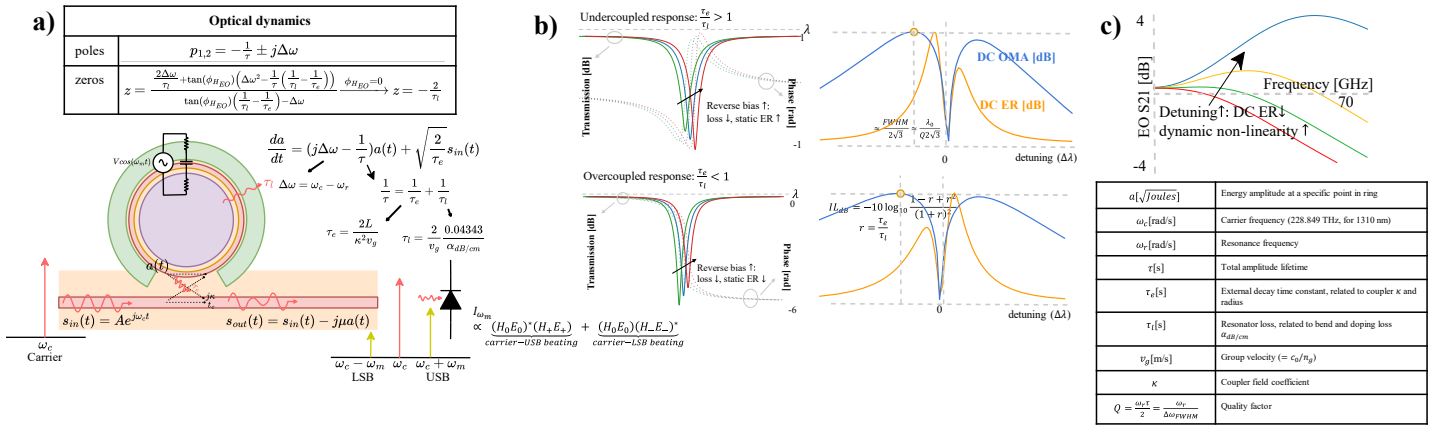


Fig. 2: Illustration of the optical response of the ring modulator at a) small signal and b) DC, and c) EO peaking trade-off

Increasing Q, or reducing the detuning, steepens the slope of the Lorentzian transfer function, enhancing intensity modulation for a given drive voltage, but a higher Q prolongs the resonator memory and reduces bandwidth. Smaller detuning can increase ER but comes with increased bias loss penalizing the optical modulation amplitude (OMA). For MRMs, the DC OMA is maximized near $\Delta\lambda \approx \lambda_0/(2\sqrt{3} \cdot Q)$ [11], independent of the coupling condition; at this point, a critically coupled modulator has 6 dB of IL, and the IL generally decreases for coupling conditions away from critical.

The detuning-ER also trades-off with bandwidth as the detuning sets the strength of the EO peaking from carrier-sideband beating; peaking is strongest when the modulation sideband aligns with the cavity resonance. The peak occurs near the same frequency as the detuning, so a smaller detuning moves the peak frequency lower. Because bandwidth depends on detuning, the DC OMA and bandwidth cannot be optimized independently [7,8,11]. A practical approach is to maximize the gain-bandwidth product, which occurs at a detuning on the order of $\Delta\omega \approx 1.13/\tau$ [7] ($\Delta\lambda = 0.565\lambda_0/Q$), with the exact optimum depending on the coupling ratio, and is generally highest for over-coupled rings. The dependence of BW on detuning results in dynamic nonlinearity, in addition to the static linearity from the Lorentzian, which is detrimental for PAM modulation formats at high baud rate, where SNR margins are already tight. Non-linear equalization is often employed to [14].

2.3. Compact modeling

These coupled electrical and optical trade-offs imply that enabling >200 Gb/s with robust performance requires co-optimization of the electronics and the photonics. At the high optical powers needed to meet receiver SNR, MRM behavior is impacted by nonlinearities coupled to each other. Recent work [15] highlights that photocurrent can couple back to resonance detuning distorting the EO response. Therefore, co-simulation with accurate electro-opto-thermal models that include self-heating, two-photon absorption (TPA), photocarriers, free-carrier dispersion, and free-carrier absorption (FCA) is crucial, and compact models make these interactions tractable in circuit simulation [16]. SPICE-compatible thermal networks that include absorption-driven self-heating and crosstalk (e.g., Foster-ladder implementations) have been recently demonstrated [17]. Other recent developments with Verilog-A models include: data driven, bidirectional photonic compact models by representing optical signals as power waves [18], nonlinear photonic interactions (e.g., four-wave mixing and cross-phase modulation) [19]. Tool vendors have also expanded

their capabilities (e.g. Spectre Photonics). Given the evolving physics, editable compact-model code is critical, as model assumptions must be iterated rapidly to incorporate new measured behavior.

3. Recent literature and comparisons

The coupled nature of the MRM design space makes benchmarking across literature inherently difficult. A community-accepted standard for biasing modulators, metric definitions, and RF de-embedding for junction extraction would improve reproducibility, but no such standard currently exists. Reported line rates also depend on instrumentation quality and DSP assumptions. Keeping that in mind, in Table 1, we list some of the reported high bitrate MRMs.

Table 1: Summary of recent MRM designs for different PN junctions:
VPN: vertical, LPN: lateral, m: meandered, Z: Z-shape, U: U-shape, CAP: SISCAP/MOSCAP

Rib Height/Slab Height/Wg Width [nm]	Junction Band	Cj [fF]	pm/V	VpiL [V-cm]	Bitrate [Gbps]	Reference
220/70/380	LPN O	10	-	0.9	290	10.1364/OFC.2025.M3K.5
-/-/380	LPN O	60	33	0.69	256	10.1364/OFC.2025.Th1E.3
-/-/-	VPN O	25	-	0.48	240	10.1364/OFC.2022.M2D.4
220/70/420	LPN O	38	-	0.82	240	10.1364/PRJ.441791
220/90/500	LPN C	10.7	30.4	-	200	10.1109/JLT.2025.3561153
220/90/500	mPN C	18.48	33.5	-	200	10.1364/OE.560256
220/90/500	ZPN O	6.6/13.2	-	0.53/0.69	200	10.1038/s41467-024-45301-3
-/-/-	VPN O	33	-	0.56	192	10.1364/CLEO_SI.2021.SF1C.3
220/90/420	LPN O	3.34	-	0.92	140	10.3390/electronics13152890
220/90/450	ZPN O	34	23	-	100	10.1109/LPT.2022.3170554

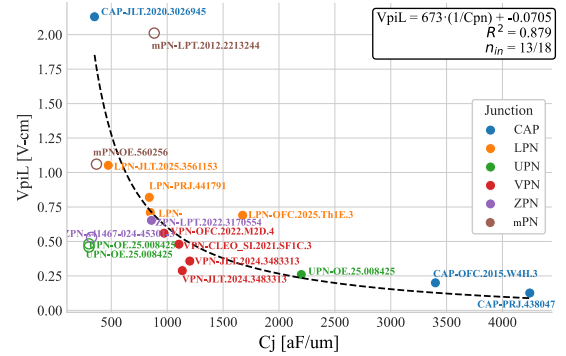


Fig. 3: VpiL-Cpn tradeoff

We focus on the phase-shifter efficiency–capacitance constraint, primarily set by depletion-region/optical-mode overlap. In Fig. 3, we plot a VpiL after scaling for radius, estimated fill factor, and our best guess for wavelength differences. Most junctions fall near a common trade-off curve, and those that are in the lower-left offer the most favorable trade-off. Because the maximum modulation amplitude in an MRM follows $1.3Q/n_g$ [8], gains in phase-shifter efficiency achieved via higher junction loss (lower Q) are offset by reduced cavity enhancement and must be balanced through the coupling coefficient.

Overall, the discussion in this paper points to holistic co-optimization of the junction, resonator coupling, and tightly integrated electronics as a requirement for scaling silicon MRMs to higher data rates.

4. References

- [1] S. Mookherjee and A. Yariv, "Coupled resonator optical waveguides," *IEEE J. Sel. Topics Quantum Electron.* 8, 448–456 (2002)
- [2] Q. Xu, B. Schmidt, S. Pradhan, et al., "Micrometre-scale silicon electro-optic modulator," *Nature* 435, 325–327–325–327 (2005)
- [3] "NVIDIA Silicon Photonics for Agentic AI Networking," <https://www.nvidia.com/en-us/networking/products/silicon-photonics/>
- [4] NVIDIA, *GTC March 2025 Keynote with NVIDIA CEO Jensen Huang* (2025)
- [5] S. K. Yeh, C. T. Shih, F. Yuan, et al., "Silicon Photonics Platform for Next Generation Data Communication Technologies," in (IEEE, 2024), pp. 1–4–1–4
- [6] W. D. Sacher and J. K. S. Poon, "Dynamics of microring resonator modulators," *Opt. Express* 16, 15741–15741 (2008)
- [7] S. Karimelahi and A. Sheikholeslami, "Ring modulator small-signal response analysis based on pole-zero representation," *Opt. Express* 24, 7585–7585 (2016)
- [8] J. Müller, F. Merget, S. S. Azadeh, et al., "Optical Peaking Enhancement in High-Speed Ring Modulators," *Sci. Rep.* 4, (2014)
- [9] H.-C. Kao, M.-W. Lin, and M.-C. M. Lee, "Study of Electrical and Optical Peaking of Si Ring Modulators for Tailoring Modulation Band," *IEEE Photonics J.* 15, 1–6–1–6 (2023)
- [10] Y. Ehrlichman, A. Khilo, and M. A. Popović, "Optimal design of a microring cavity optical modulator for efficient RF-to-optical conversion," *Opt. Express* 26, 2462 (2018)
- [11] H. Yu, D. Ying, M. Pantouvaki, et al., "Trade-off between optical modulation amplitude and modulation bandwidth of silicon micro-ring modulators," *Opt. Express* 22, 15178 (2014)
- [12] C. Xie, M. Raj, A. Joshi, et al., "A 64 Gb/s NRZ O-Band Ring Modulator with 3.2 THz FSR for DWDM Applications," in *2024 Optical Fiber Communications Conference and Exhibition (OFC)* (2024), pp. 1–3
- [13] M. F. Chen, H. T. Cheng, C. H. Tsou, et al., "Optical and Electrical Characterization of a Compact Universal Photonic Engine," in (2025), pp. 54–58
- [14] H. Li, G. Balamurugan, T. Kim, et al., "A 3-D-Integrated Silicon Photonic Microring-Based 112-Gb/s PAM-4 Transmitter With Nonlinear Equalization and Thermal Control," *IEEE J. Solid-State Circuits* 56, 19–29 (2021)
- [15] N. Nedovic, N. Mehta, A. Rekhi, et al., "Carrier Photogeneration Impact on Ring Dynamics," in *2025 IEEE Photonics Society Summer Topicals Meeting Series (SUM)* (IEEE, 2025), pp. 1–2
- [16] M. De Cea, A. H. Atabaki, and R. J. Ram, "Power handling of silicon microring modulators," *Opt. Express* 27, 24274 (2019)
- [17] D. Coenen, M. Kim, and H. Oprins, "Electro-Opto-Thermal Dynamic Compact Model for Ring Modulator Arrays," *IEEE Trans. Compon., Packag. Manuf. Technol.* 15, 1006–1013 (2025)
- [18] D. Azhigulov, Z. Lu, J. Pond, et al., "Enabling data-driven and bidirectional model development in Verilog-A for photonic devices," *Opt. Express* 32, 29965 (2024)
- [19] S. Zhang, X. Fan, N. Chen, et al., "Compact photonic model based on coupled-mode theory for nonlinear interactions in electronic-photonic co-simulation," *Opt. Express, OE* 32, 27599–27613 (2024)